Answer Key & Teacher Notes

Session 1 — Independent and Dependent Variables · Session 2 — Tables of Values

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	About 10 minutes; use items 1–4 for a 5-minute check. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 · TEACHER NOTES

Independent and Dependent Variables

TALK PROMPTS

- Can the same quantity be independent in one pair and dependent in another pair? How?
- Could the weather also be an independent variable in the popcorn story? What would it change?

PARTNER TASK

Charging a Phone. Elena's battery percent rises the longer her phone is plugged in. One partner names the independent and dependent variables; the other checks with the "depends on" sentence. Switch with Mowing Lawns: lawns mowed and gas used.

WATCH FOR

- Choosing a variable by the size of its numbers (the bigger number is dependent).
- Reversing the sentence test ("drinks sold depend on money collected").
- Thinking a quantity is always independent or always dependent, no matter the pair.

SESSION 2 · TEACHER NOTES

Tables of Values

TALK PROMPTS

- Why does the independent variable belong in the first column of a table?
- How can you check that every row of your table follows the same rule?

PARTNER TASK

Bracelet Beads. One partner picks three numbers of bracelets; the other finds the beads with the rule $y = 24x$. Then switch, and each partner checks one row by multiplying.

WATCH FOR

- Adding the fee before multiplying ($5 + 3 = 8$, then 8×4).
- Putting the dependent variable in the first column, as in Tavon's train table.
- Adding the rate between rows when the inputs skip (x: 2, 4, 6 means y grows by 2 times the rate).

ITEM ANALYSIS

suggested DOK; judge the reasoning required, not the question label

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 · Form A	6.EE.C.9	1	2	1	2	1	1	2	3	2	1	1
S1 · Form B	6.EE.C.9	1	2	1	2	1	1	2	3	2	1	1
S2 · Form A	6.EE.C.9	1	2	1	1	2	2	2	3	2	1	1
S2 · Form B	6.EE.C.9	1	2	1	1	2	2	2	3	2	1	1

Items before the last three are Practice; then Find the Error (explain and correct) and two Test Practice items.

GUIDED NOTES

independent variable: The **input**: the quantity you choose or that comes first.

dependent variable: The **output**: it changes because of the input.

depends on: The output **changes** when the input changes.

- Sentence test: the dependent variable **depends on** the independent variable.
- On a graph, the independent variable goes on the **x-axis** and the dependent variable goes on the **y-axis**.

Example: The two quantities are hours rented and **total cost**. The cost changes when the hours change, so cost **depends on** hours. Answer: Independent: **hours** · Dependent: **cost**

YOU TRY

1. a. calories burned b. total bill

2. The number of levels completed depends on the time Rosa plays.

EXIT TICKET

1. a. hours worked b. hours studied

2. a. pounds bought b. cost

3. **B** (ice creams sold)

PRACTICE FORM A

1. a. hours worked I, money earned D b. test score D, hours studied I c. miles driven I, gallons of fuel D d. guests at a party I, pizzas needed D

2. a. Independent: bags sold; dependent: money collected b. Independent: game length; dependent: bags sold c. bags sold

3. a. watering (how often Theo waters) b. height (how tall each fern grows)

4. Disagree. The riders depend on the distance traveled, so distance is independent and riders are dependent.

5. Dependent: money earned (it depends on the hours walked).

6. a. miles (independent) b. cost (dependent)

7. The download time depends on the number of photos. Dependent: time (photos = independent).

8. Disagree. The size of a number does not decide. Money collected depends on pairs sold, so pairs sold is independent and money collected is dependent.

9. She switched the variables. Selling drinks causes the money, so: "The money collected depends on the number of drinks sold." Drinks sold is independent; money is dependent.

10. **A** (*x*-axis)

11. **B** (money earned)

Self-check word **OUTPUT**

PRACTICE FORM B

1. a. pages read D, minutes reading I b. cups sold I, money collected D c. calories burned D, minutes jogged I d. players on a team I, water bottles needed D

2. a. Independent: cups sold; dependent: money collected b. Independent: temperature; dependent: cups sold c. cups sold

3. a. light (hours of light each plant gets) b. leaves (how many leaves grow)

4. Agree. The water in the pool depends on the time the hose runs, so time is independent and water is dependent.

5. Dependent: gas used (it depends on the lawns mowed).

6. a. hours (independent) b. height (dependent)

7. The battery percentage depends on the time plugged in. Dependent: battery (time = independent).

8. Disagree. The size of a number does not decide. Money collected depends on muffins sold, so muffins sold is independent and money collected is dependent.

9. He switched the variables. Practice time causes the score, so: "Her free-throw score depends on the minutes she practices." Minutes is independent; score is dependent.

10. **B** (*y*-axis)

11. **B** (gallons of fuel)

Self-check word **DEPEND**

GUIDED NOTES

table of values: A list of **inputs** matched with their outputs.

input: The number you start with. It goes in the **first** column.

output: The number you get **after** you use the rule.

- Use the **same** rule on every input.
- With a fee, **multiply** first, then add the fee.

Example: $h = 1$: $9 \times 1 = \mathbf{9}$ $h = 2$: $9 \times 2 = \mathbf{18}$ $h = 3$: $9 \times 3 = \mathbf{27}$ Answer: Pairs: (1, 9), (2, 18), (3, **27**)

YOU TRY

1. 1, 6, 11, 16

2. a. 70 pages b. 140 pages

EXIT TICKET

1. 4, 12, 24

2. $2 \times 3 + 5 = \$11$

3. **A** (first column)

PRACTICE FORM A

1. a. 30 b. 14 c. 16 d. 100

2. 30, 60, 90, 150; 8 hours: $30 \times 8 = 240$ pages

3. 36, 54, 90; cartons (y) go on the vertical axis

4. $25 \times 6 = 150$ cents

5. No. Cars depend on crossings, so crossings (x) go in column 1 and cars (y) go in column 2.

6. \$8, \$11, \$17; 6 hours: $3 \times 6 + 5 = \$23$

7. a. Multiply x by 12 ($y = 12x$) b. $12 \times 7 = 84$

8. Plan A: 6, 12, 18, 30. Plan B: 14, 18, 22, 30. 8 hours: Plan B, \$42 (Plan A is \$48).

9. He switched the columns. Miles depend on hours, so hours go in the x column: x: 1, 2, 3 and y: 60, 120, 180.

10. **B** (30)

11. **A** (14)

Self-check word **TABLE**

PRACTICE FORM B

1. a. 42 b. 19 c. 20 d. 60

2. 24, 48, 72, 120; 8 bracelets: $24 \times 8 = 192$ beads

3. 18, 30, 42; slices (y) go on the vertical axis

4. $7 \times 8 = 56$ cups

5. Yes. Bikes depend on sections, so sections (x) go in column 1 and bikes (y) go in column 2.

6. \$10, \$14, \$22; 7 hours: $4 \times 7 + 6 = \$34$

7. a. Multiply x by 15 ($y = 15x$) b. $15 \times 8 = 120$

8. Plan A: 8, 16, 24, 40. Plan B: 17, 22, 27, 37. 8 hours: Plan B, \$52 (Plan A is \$64).

9. He switched the columns. Miles depend on hours, so hours go in the x column: x: 1, 2, 3 and y: 50, 100, 150.

10. **B** (32)

11. **C** (13)

Self-check word **INPUT**

SHEET 1 · INDEPENDENT AND DEPENDENT VARIABLES

1. a. minutes biking I, miles traveled D b. total cost D, tickets bought I c. guests I, cups of juice D
2. Model: "Her distance from home depends on the time she rides the bus." (time riding = independent, distance from home = dependent)
3. Independent: number of dogs walked. Dependent: money earned (it depends on the number of dogs).
4. a. weeks (on the x-axis) b. height in inches (on the y-axis) c. After 4 weeks, the sunflower is 20 inches tall.
5. The size of a number does not decide the variable. The sandwiches needed depend on the number of people, so the number of people is independent and the number of sandwiches is dependent.

SHEET 2 · TABLES OF VALUES

1. 8, 10, 12, 15
2. $8, $11, $14, $23
3. Multiply the boxes by 8 (y = 8x); 4 boxes: 32 markers, 8 boxes: 64 markers
4. The last value, 8, is wrong. When x = 4, y = 2 × 4 + 1 = 9 (the y values go up by 2 each time).
5. $6, $10, $14; $20 is 8 miles (2 × 8 + 4 = 20)
6. **C** (29)
7. **B** (7, 8, 9)

October 2026 conceptual additions — teacher key

Session 1 · Independent and Dependent Variables

Length bought is input/independent; total cost is dependent on length. Costs $0,$6,$15. Variables must be identified from the model, not from their numerical size.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: Length bought is input/independent; total cost is dependent on length. Costs $0,$6,$15. Variables must be identified from the model, not from their numerical size.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.

Session 2 · Tables of Values

Outputs 5,9,15. The $5 fee appears at zero boxes. Input jumps 2 then 3, so outputs increase 4 then 6; per-box rate is 2.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: Outputs 5,9,15. The $5 fee appears at zero boxes. Input jumps 2 then 3, so outputs increase 4 then 6; per-box rate is 2.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.